

Luciteria Collector Cabinet — Production Roadmap

What the Prototype Proves

Phase 1 (Complete)

1.  Collection tracking UX works for element collectors
2.  Assignment engine logic prevents duplicates across 5 modes
3.  Admin operations dashboard provides full visibility
4.  Customer preference system is expressive yet simple
5.  Edge cases (complete collection, empty collection, low stock) are handled
6.  Brand voice integrates naturally into UI microcopy

Phase 2 (Complete)

1.  Interactive periodic table (118 elements, standard layout, click-to-detail)
2.  Collection type system (5 types, onboarding, per-customer tracking)
3.  Enhanced assignment engine (collection type filtering, precious metal exclusion, discount tracking, sequence preview, OOS shifting)
4.  Billing system (dual billing model, 5-day edge case, grandfathering with pause clock)
5.  Discount monitoring (>20% admin alerts, per-shipment tracking)
6.  White/light theme (Luciteria branding, mobile responsive)
7.  Notification stubs (8 event types, console/DB logging)
8.  Expanded data (35 products, 6 customers, all collection types)
9.  Forward-compatible schema for multi-collection support

What Needs to Change for Production

Phase 3: Shopify Integration (2-3 weeks)

Authentication & App Bridge

- ☐ Register app in Shopify Partners dashboard
- ☐ Implement OAuth flow with `@shopify/shopify-app-remix`
- ☐ Configure App Bridge for embedded admin experience
- ☐ Set up session management and token storage

Shopify API Integration

- ☐ Replace mock product data with Shopify GraphQL Admin API
- ☐ Sync products via `products` query with metafields
- ☐ Map Shopify customers to internal Customer records
- ☐ Listen to order webhooks to auto-populate CollectionRecords
- ☐ Use Shopify's inventory API for real-time stock levels

Data Model Migration

- ☐ Move from SQLite to PostgreSQL (or Shopify's built-in storage)

- [] Add Shopify metafield mappings for element metadata
- [] Create product metafields: `elementSymbol`, `atomicNumber`, `rarityTier`, `collectionType`
- [] Customer metafields: `collectorProfile`, `subscriptionPreferences`, `collectionType`

Phase 4: Subscription & Billing Production (2-3 weeks)

Shopify Subscriptions API

- [] Integrate with Shopify Subscription APIs (selling plans)
- [] Create subscription selling plans per collection type
- [] Handle subscription lifecycle events (create, pause, cancel, skip)
- [] Billing integration via Shopify Billing API
- [] Implement production dual billing model (replace stubs)
- [] Implement grandfathering with real payment provider

Assignment Automation

- [] Move assignment engine to a scheduled job (cron / Shopify Flow)
- [] Run assignments X days before billing date
- [] Auto-create draft orders for approved assignments
- [] Send assignment preview emails to customers
- [] Implement inter-subscriber conflict resolution for low-stock items

Phase 5: Customer Experience (1-2 weeks)

Storefront Integration

- [] Build customer-facing pages using Shopify Customer Account Extensions
- [] Or: create a standalone portal accessible from customer account
- [] Add “My Collection” link to Shopify theme navigation
- [] Implement real add-to-cart for wishlist items
- [] Real-time inventory display

Notifications (Replace Stubs)

- [] Email templates (SendGrid/Postmark) for all 8 notification types
- [] SMS integration (optional, via preferences)
- [] Shopify Flow triggers for collection milestones
- [] Webhook integrations for external systems

Multi-Collection Support

- [] Allow customers to subscribe to multiple collection types simultaneously
- [] Separate billing/assignment per collection type
- [] Combined dashboard view across all collections
- [] Collection type switching with proper billing adjustments

Phase 6: Admin & Operations Hardening (1-2 weeks)

Admin Hardening

- [] Role-based access control (replace session-based prototype auth)
- [] Audit log for all admin actions (assignments, overrides, notes)
- [] Bulk operations (assign multiple, skip multiple)
- [] Export reports (CSV of assignments, collection status, discount analysis)

Assignment Engine Production Features

- ☐ Multi-subscriber conflict detection (shared low-stock items)
- ☐ Machine learning recommendations (predictive preferences)
- ☐ A/B testing different assignment strategies per cohort
- ☐ Production discount alerting (email to admin, not just UI badge)

Phase 7: Polish & Launch (1-2 weeks)

UI/UX

- ☐ Replace inline styles with proper CSS modules
- ☐ Add loading states, skeleton screens, error boundaries
- ☐ Accessibility audit (WCAG 2.1 AA)
- ☐ Product image integration from Shopify CDN
- ☐ Periodic table performance optimization for mobile

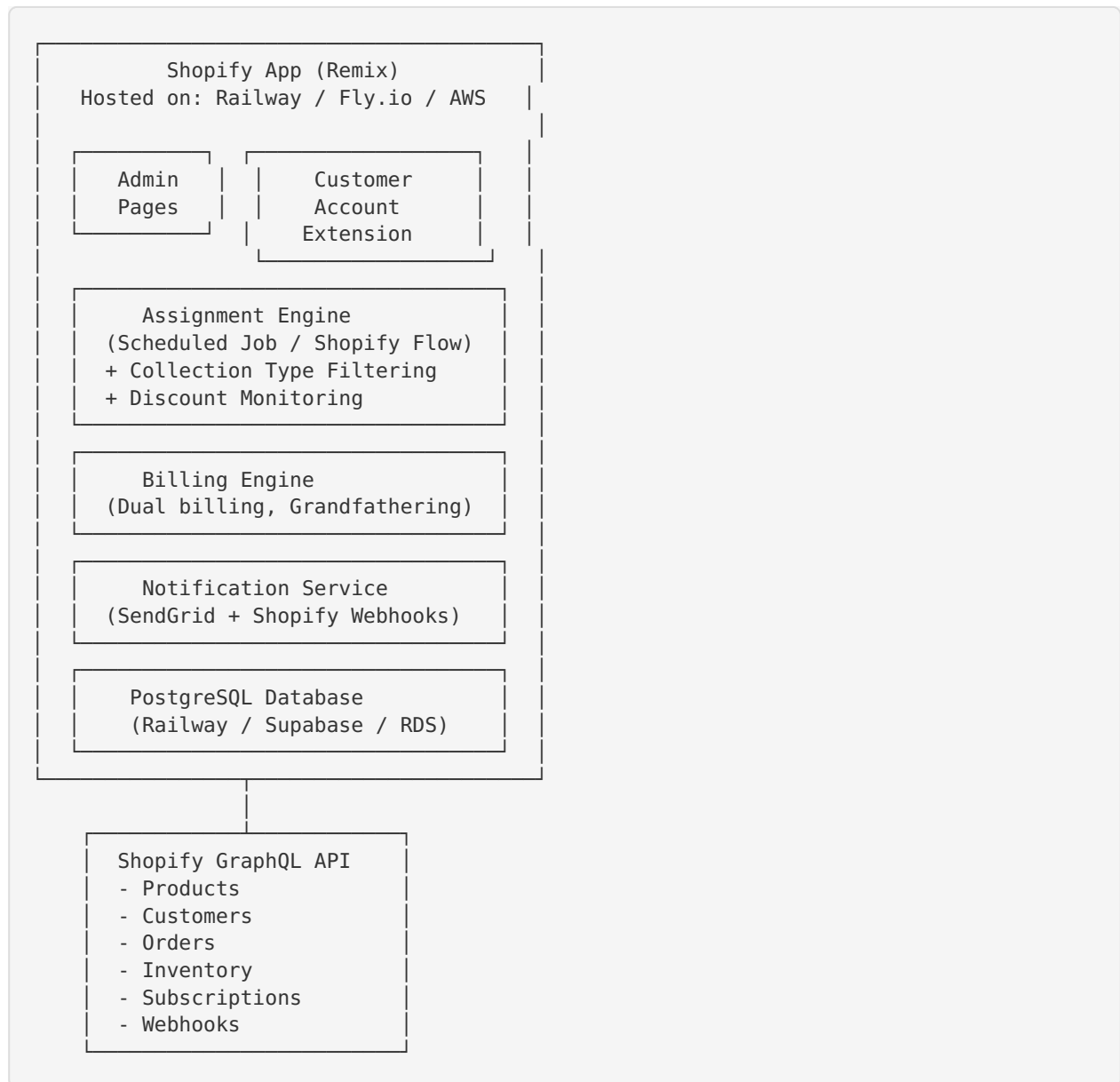
Performance

- ☐ Database query optimization and indexing
- ☐ Caching for product catalog and inventory
- ☐ Rate limiting for API calls
- ☐ Background job processing for assignments

Testing

- ☐ Unit tests for assignment engine (all strategies, all modes, all collection types)
- ☐ Integration tests for Shopify API interactions
- ☐ End-to-end tests for customer flows
- ☐ Load testing for assignment engine at scale
- ☐ Billing edge case regression tests

Production Architecture



Estimated Timeline

Phase	Duration	Dependencies
Phase 1: Core Prototype	✓ Complete	—
Phase 2: Enhanced Prototype	✓ Complete	—
Phase 3: Shopify Integration	2-3 weeks	Shopify Partners account, API keys
Phase 4: Subscription & Billing	2-3 weeks	Phase 3 complete
Phase 5: Customer Experience	1-2 weeks	Phase 3 complete
Phase 6: Admin Hardening	1-2 weeks	Phase 4 complete
Phase 7: Polish & Launch	1-2 weeks	All phases complete
Total remaining	7-12 weeks	

Cost Estimates

Service	Monthly Cost
Hosting (Railway/Fly.io)	\$20-50
PostgreSQL (managed)	\$15-30
Shopify App (custom)	\$0 (own store)
Email service (SendGrid/Postmark)	\$0-20
Monitoring (Sentry)	\$0-26
Total	\$35-126/mo